ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **WINTER SEMESTER, 2021-2022**

DURATION: 1 HOUR 30 MINUTES **FULL MARKS: 75**

CSE 4107: Structured Programming I

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

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1. a) The double factorial of a number n, denoted by n!!, is the product of all the integers from 1 up to n that have the same parity (odd or even) as n. 15 (CO3) (PO2)
For example:
Double factorial of 5!! = 5 × 3 × 1 = 15
Double factorial of 6!! = 6 × 4 × 2 = 48
Solve the problem of finding the double factorial of a number n, by writing a C program.

b) Modify the following lines of codes given in the Table 1 by rearranging them, so that it produces the expected output. 10 (CO3) (PO2)

Table 1: Sample program and expected output

Given Code	Expected Output

`#include<stdio.h>`
`int main(){`
`int i, j;`
`j = (i/10);`
`for(; i > 0; i--)`
`for(i=100; i>=0; i-=10){`
`for(i=j; i<5;i++)`
`if(j%2)`
`printf("%d ",j);`
`printf("%d ",2*i-1);`
`printf("%d ",9-2*i);`
`printf("\n");`
`printf("\n");`
`return 0;`
`}`
`}`
	9 7 5 3 1
	9 7 5 3 1
	9 7 5 3 1

[NB: You may only rearrange the statements, do not add, remove, or modify any lines of the given code]

2 a) Write the output for each of the code segments provided below: 7.5 (CO1) (PO1)

i.) `printf("\\/\\/\n/\\/\\\n");`

ii.) `printf("%%%d%c%d=%c%%\n",3,'+',2,'?');`

iii.)
```
int a=4, b=1;
printf("%d %d\n", a++, ++b);
```

iv.)
```
int i=37;
printf("%d\n",(i%10)*10 + (i/10));
```

v.)
```
char ch='A';
printf("%c %d %d\n", ch+1, ch+1, ch= ='A');
```

b) Find whether the following conditions are true or false. Also find the updated values of the variable 7.5

after evaluating the expression. (CO1)

i.) if(a++ && --b || --c) (PO1)

ii.) if(++a && b-- || --c)

iii.) if(c-- || ++a && b--)

iv.) if(--c || ++a && b--)

v.) if(a = 0 || (b=10) && (c=12))

For each of the case, consider the variables are initialized as below:

```
int a = 0, b = 1, c = 1;
```

[NB: 50% marks will be deducted for each wrong answer]

c) Answer the following questions: 10

i.) Why is it important for the *scanf* function to know the address of the variable, but not for the *printf* function? (CO1) (PO1)

ii.) Why is it necessary to allocate consecutive memories for the elements of an array?

iii.) Write short notes on typecasting with an example.

iv.) Can you find a value of the variable 'x', so that the condition **`if(30 < x < 40)`** becomes false? If your answer is yes, find the value. If your answer is no, explain why.

v.) Write down the basic data types and their size in the memory.

3. a) Suppose you want design an application, that asks the user to enter a string as an input and prints the occurrence of the last character in the string. Solve the problem by writing C program. 12 (CO3) (PO2)
For example:
If user provides "programming" as an input, the program should print 2 as an output, as the last character 'g' appears twice in the string.

b) Examine the following program and expected output provided in the Table 2. Due to logical errors, the program was not producing the expected output. 5+8 (CO2) (PO1) (CO3) (PO2)

Table 2: Sample program and expected output

Program	Expected Output

`#include<stdio.h>`
`int main(){`
`  int num=12345678, sum, rem;`
`  while(num > 0){`
`  num /= 10;`
`  rem = num%10;`
`  if(rem%2){`
`  printf("%d + ", rem);`
`  sum += rem;`
`  }`
`  }`
`  printf("0 = %d",sum);`
`  return 0;`
`}`
	8 + 6 + 4 + 2 + 0 = 20

Answer the following questions:

i.) Find the output of the given program.

ii.) Modify the program with minimum changes so that it produces the expected output.

[NB: Any unnecessary changes will cause penalty]